Natural Selection: Heterozygote Advantage

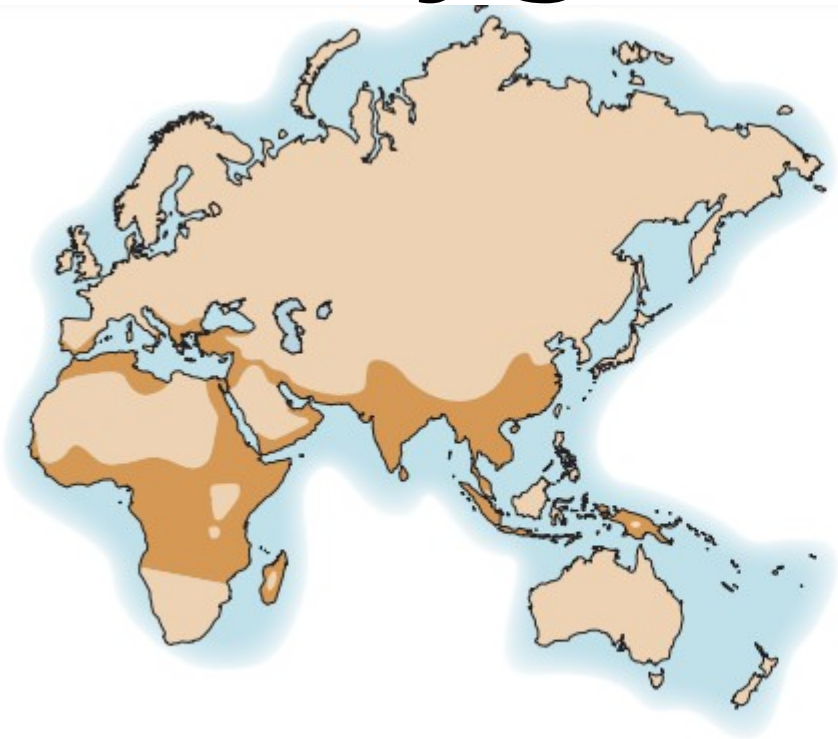


Figure 14.12 The distribution of malaria

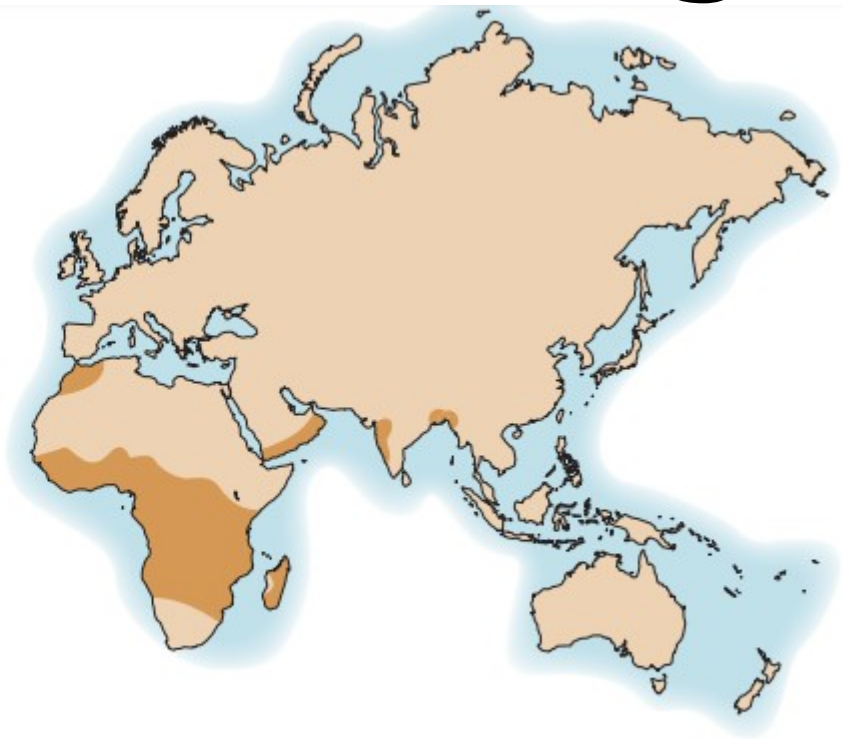


Figure 14.13 The distribution of sickle-cell anaemia

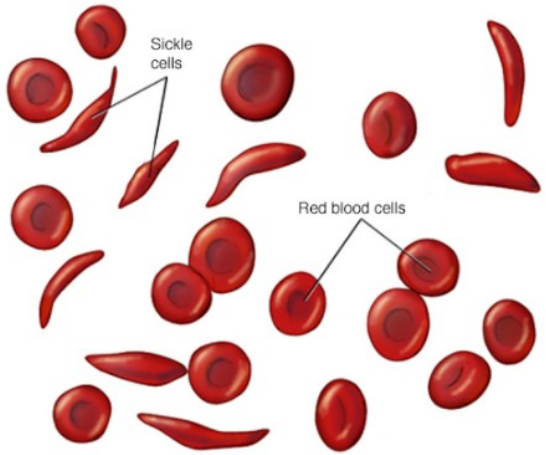
Heterozygote Advantage

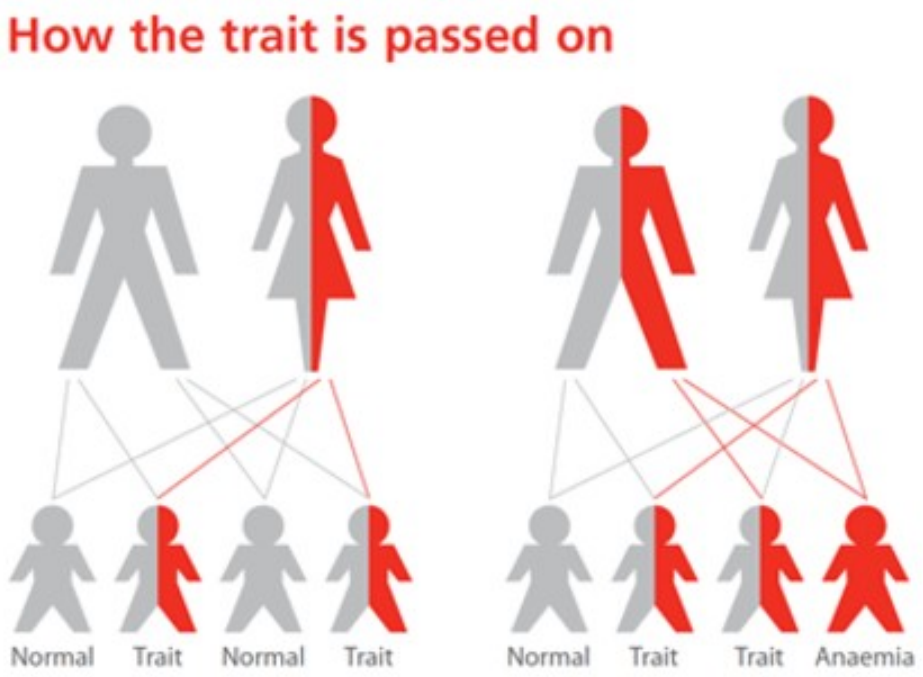
- Heterozygote Advantage is when carrying one allele for a recessive genetic disease is an advantage in a particular environment.
  - Carriers have a survival advantage.
  - Allele frequency rises even though there is disadvantage for homozygous recessive people who get the disease.
- Examples:
  - Sickle Cell Anaemia
  - Alpha Thalassemia
  - Tay-Sachs Disease

Learning Aim: Define heterozygote advantage and give examples

Heterozygote Advantage Example 1: Sickle Cell Anaemia

- Varies in incidence throughout the world.
- Faulty allele caused by base substitution.
- Allele causes “sickle” shape of RBC.
- If only one allele inherited (heterozygous):
  - Only some blood cells are sickled
  - No symptoms, unless extreme conditions of lower O_2
  - Carrier of allele – can be passed on to offspring.
- If two alleles are inherited (homozygous):
  - All blood cells affected:
  - Faulty haemoglobin
  - Inability to adequately transport oxygen
  - Death in childhood
- Two carriers must reproduce to have a 25% chance of a child being homozygous for sickle cell anaemia.





Learning Aim: Discuss causes, inheritance and symptoms of sickle-cell anaemia

So when is carrying one allele for sickle cell anaemia an advantage?

- Using our understandings of natural selection, we would expect the sickle cell allele to decrease in the population over time.
- This has happened in most locations; *except in areas where there is malaria*

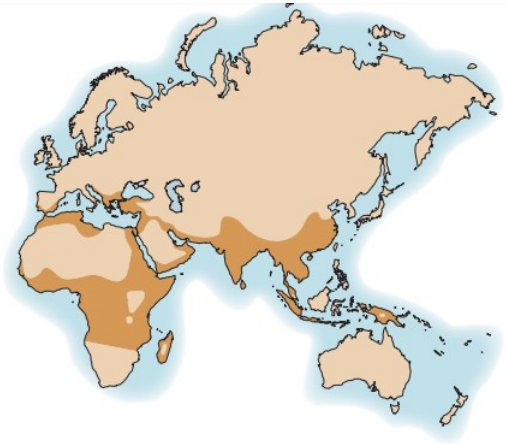


Figure 14.12 The distribution of malaria

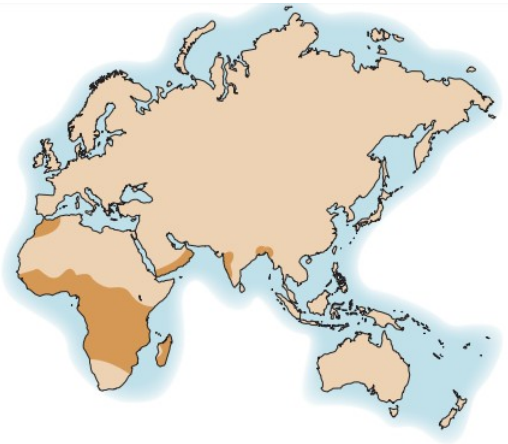


Figure 14.13 The distribution of sickle-cell anaemia

- There is greater incidence of sickle cell anaemia in areas where malaria is also present because:

Carrying the allele for sickle-cell anaemia protects carriers against malaria

- In these areas of the world:
  - People who inherit two copies of the allele die.
  - People who inherit one copy are resistant to malaria as a result.
  - People who inherit no copies (normal blood) are at greater risk of dying from malaria.
- Carrying one copy of the sickle cell allele is an advantage compared to carrying no copies or two copies, so the allele frequency is higher in areas where malaria is present.

Learning Aim: Explain why allele frequency for sickle cell anaemia rises in areas where malaria is present

So when is carrying one allele for sickle cell anaemia an advantage?

- Areas with no malaria:
  - greatest advantage is held by individuals with normal blood (no sickle cell allele): they are most likely to survive and bear young.
- Areas with malaria:
  - people with normal blood survive to bear young, but some die from malaria.
  - People who carry one sickle cell allele survive almost as well: malaria resistance offsets the disadvantage from the sickling allele for carriers.
  - Therefore the allele is passed on to a greater degree than in areas without malaria.

Presence of malaria acts a selection pressure for the sickle cell allele

Learning Aim: Explain why allele frequency for sickle cell anaemia rises in areas where malaria is present

Heterozygote Advantage Example 2: Alpha (α) Thalassemia

- Autosomal recessive disease causing defects to haemoglobin production. Often caused by base insertion.

- Haemoglobin is made of four proteins.

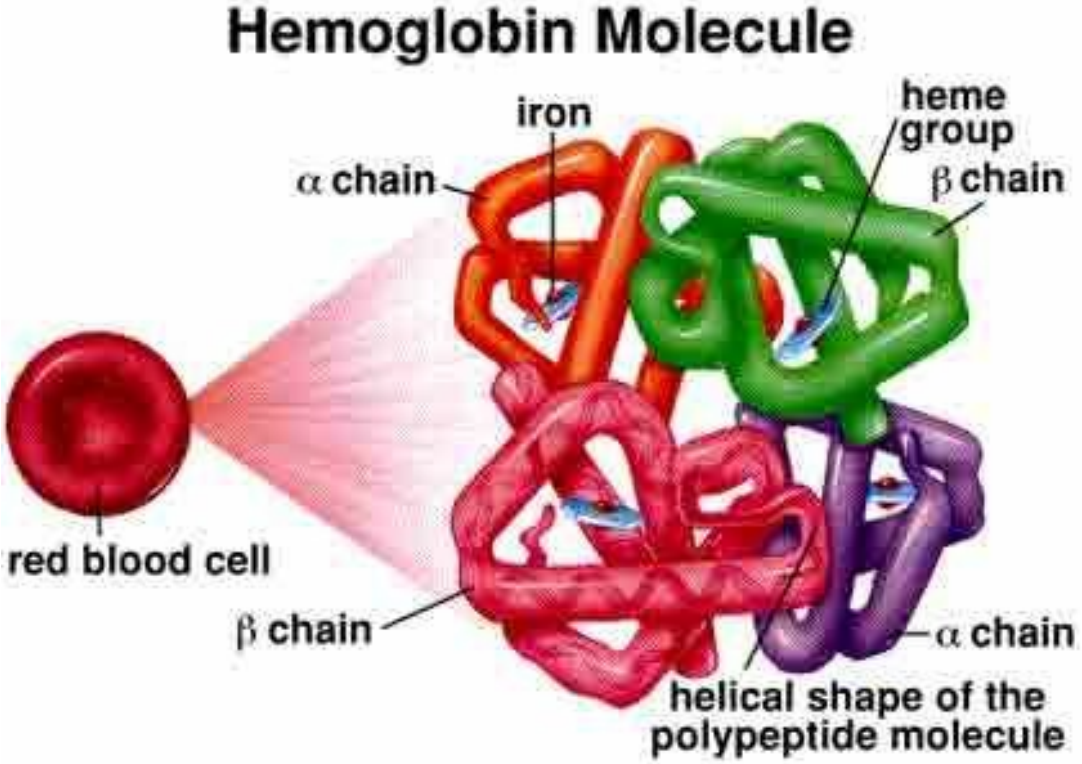
- Two types of thalassemia:

Alpha Thalassemia -

- defects to alpha globin
- Heterozygote advantage where malaria is present
- Symptoms of anaemia and sometimes high iron levels

Beta Thalassemia – defects to beta globin

- Defects to beta globin
- No evidence of heterozygote advantage
- Symptoms of anaemia and sometimes high iron levels



Learning Aim: Discuss causes, inheritance and symptoms of alpha thalassemia

Heterozygote Advantage Example 2: Alpha Thalassemia

- Alpha Thalassemia is far more common in areas where malaria is present.

- People who are heterozygous for Alpha Thalassemia are less likely to get malaria, or will experience milder symptoms.
  - Homozygous Dominant: don't get alpha thalassemia but are more likely to get severe malaria
  - Heterozygous:
    - get mild thalassemia but mild or no malaria
    - this has the best survival in areas with malaria – it is a selective advantage.
    - due to the survival advantage, people who are heterozygous are more likely to survive and reproduce – so the allele frequency is higher in populations with malaria.
  - Homozygous recessive: die at birth



Learning Aim: Discuss causes, inheritance and symptoms of alpha thalassemia

Heterozygote Advantage Example 2: Alpha Thalassemia

Alpha (α) Thalassemia

- Defects in alphaglobin protein structure in haemoglobin cause anaemia
- More common in South East Asia
- Autosomal recessive inheritance
- 2 gene loci involved (4 alleles) on chromo 16
- Faulty allele for alphaglobin
- Homozygous Dominant
  - no faulty alleles, no anaemia
- Heterozygous
  - some faulty alleles - mild to moderate anaemia
- Homozygous recessive
  - all faulty alleles – fatal at birth
- **Heterozygote advantage where malaria is present**

Beta (β) Thalassemia

- Defects in betaglobin protein structure in haemoglobin cause anaemia
- More common in the Mediterranean Basin
- Autosomal recessive inheritance
- 1 gene locus involved (2 alleles) on chromo 11
- Faulty allele for betaglobin
- Homozygous Dominant
  - No faulty allele, no anaemia
- Heterozygous
  - One faulty allele, mild anaemia
- Homozygous recessive
  - Two faulty alleles, moderate to severe anaemia.
- **No evidence of heterozygote advantage where malaria is present**

Learning Aim: Contrast causes of increased allele frequency for alpha and beta thalassemia

Beta Thalassemia – no evidence for heterozygote advantage

- Beta thalassemia is more common in countries around the Mediterranean Basin, but there is no evidence for heterozygote advantage.

So why is the allele frequency for beta thalassemia greater around the Mediterranean?



- Due to (historically) restricted gene pool with sociocultural barriers to gene flow.
- Mutation to cause beta thalassemia occurred at some time in the past, in that area.
- Cultures in the Mediterranean often had cousin marriage to maintain power and wealth, therefore a barrier to gene flow.
- Due to breeding within a closed population, the alleles persist at a higher frequency than in other populations.

Learning Aim: Contrast causes of increased allele frequency for alpha and beta thalassemia

Heterozygote Advantage Example 3: Tay Sachs Disease

- Lethal recessive trait
- Faulty allele for production of enzyme hexosaminidase, involved in lipid metabolism

Inheriting no faulty alleles (homozygous dominant)

- Enzyme is correctly produced, no disease

Inheriting one faulty allele (heterozygous)

- No disease as the working allele produces the hexosaminidase.
- Allele can be passed to offspring
- *Heterozygote advantage in areas where tuberculosis is present*

Inheriting two faulty alleles (homozygous recessive)

- Buildup of lipids called gangliosides in the brain
- Progressive deterioration of CNS activity
- Death in infancy or early childhood

In most populations, the most advantageous genotype is to be carrying no alleles for Tay-Sachs. For this reason, the allele decreases in the population over time.

In communities with tuberculosis, people heterozygous for Tay Sachs disease are less likely to get tuberculosis. In this case, being heterozygous is a survival advantage.

In small populations affected by tuberculosis, the allele frequency will rise in the population.

Learning Aim: Discuss causes, inheritance and symptoms of Tay-Sachs Disease, and explain why allele frequency may rise in areas where Tuberculosis is present